Crop Shift Mitigation Web Platform for Promoting Oilseed Farming

Gangadhara
Dept. of CSE (Data Science)
Presidency University
Bengaluru, India
aryagangadhar09@gmail.com

Chandan Kumar H H
Dept. of CSE (Data Science)
Presidency University
Bengaluru, India
hhc90809@gmail.com

Navyashree E
Dept. of CSE (Data Science)
Presidency University
Bengaluru, India
navyashree23519@gmail.com

Dr. Radha K
Assistant Professor
Dept. of CSE
Presidency University
Bengaluru, India
(Corresponding Author)

***Abstract*— The country of India is still heavily reliant on imported edible oils because of the lack of domestic production. Among the key factors that have contributed to this imbalance is the unwillingness of farmers to embrace oilseed farming. Farmers tend to stick with old crops due to familiarity, the stable demand, and less perceived risk. Nevertheless, this model restricts the diversification of agricultural production and has an impact on national self-sufficiency. This paper will come up with a web-based decision support system to help farmers decide on the right crops to plant depending on the forecasted yield and profitability. The system combines environmental factors of rainfall, temperature, soil type, and irrigation factors with past agricultural information. The estimation of the crop yield is done by a machine learning model, namely the Random Forest model.**

Market prices are further analysed to determine profitability with the predicted yield. The system displays findings in a user-friendly format enabling farmers to make comparison of various crop alternatives. The results show that oil seed crops have the potential to yield more when conditions are favorable. This system assists in the minimization of uncertainty in the choice of crop and aids in the shift to oilseed farming.

Keywords*— *Crop Shift, Oilseed Farming, Machine Learning, Random Forest, Profitability Analysis, Decision Support System

I. INTRODUCTION

India stands as one of the largest manufacturers of oilseeds in the world, but it imports almost half of its edible oil needs, and it spends more than Rs. 18 billion a year on imports [1]. The local production is approximately 12 million tonnes as compared to demand of more than 27 million tonnes and as a result there has been a consistent shortage in supply that puts pressure on the foreign exchange reserves and the economy is prone to international price increases [4]. The gap that requires filling is the need to grow more domestic oilseeds.

Policymaking efforts notwithstanding, the farmers of oilseed deficit states still prefer the staple crops, paddy and wheat. According to field surveys, the main discouraging factors to oilseeds adoption are the uncertainty of the yields, the fluctuation of the prices of sunflower and groundnut, and the lack of location-specific advisory tools. In the absence of sound and data-supported advice, smallholders believe that the transition is too risky than standard crop cycles.

Recent studies in the field of precision agriculture have shown that machine learning models that have been trained using soil, weather, and past yield data can produce crop-specific estimates of yield with moderate accuracy [2][3]. Nevertheless, an overview of the current platforms shows that there is a fatal weakness: profitability is seldom calculated in parallel with yield, although net income and not output volume per se determine decision-making by farmers.

In an attempt to fill this gap, this work develops a web-based crop decision support system, which combines a Random Forest yield predictor and profit calculation engine. The system takes as inputs those furnished by the farmers, including land area, type of soil, source of irrigation, and season, and yields a list of comparison of crops in terms of net profit, plus a risk rating, thus transforming raw predictions into practical cultivation advice.

II. LITERATURE REVIEW

Khaki and Wang [2] established that deep neural networks using multi-year yield records in the US Corn Belt had lower error in prediction compared to classical regression, which is an advantage of non-linear models when dealing with heterogeneous agricultural data. A combination of Random Tree, K-Nearest Neighbour and Naive Bayes classifiers to recommend crops in precision agriculture was proposed by Pudumalar et al. [3], which achieved higher accuracy than any baseline model. The theoretical background of the idea of Random Forests is the work of Breiman [7], who defined the concept of variance-reducing ensembles and demonstrated their resilience to the lack of sensor data and the distortion of field measurements. The concept of Random Forests has been since broadly used in agricultural studies because of its ability to withstand the absence of sensor data and noisy field measurements.

Government-connected portals and agri-fintech platforms have facilitated access to weather alerts and mandi prices and subsidies on the advisory front to farmers. But these services are vertically siloed: yield forecasting services are not connected with market pricing services, and are not connected with risk-categorised profit estimates. Padhan [4] actually noted that the yield-based approach to the analysis of oilseed production in India cannot explain the variation in the input costs leading to inaccurate conclusions on the viability of the farm.

The review indicates that there is no current open platform that integrates location-sensitive yield forecasting and input-adjusted profit estimation and risk labeling within a single user interface focused on oilseed promotion. The current mechanism aims at addressing this gap, and the clear policy goal will be to curb the dependence of India on importing edible oil, by providing evidence-based crop switching assistance to the farmers.

III. PROPOSED SYSTEM

The suggested system is an online app that will help farmers to choose the appropriate crops based on data-driven information. The site gives the user an opportunity to enter general information like land size, the geographical position, and crop choice.

There are three key components of the system architecture:

1. **Frontend Interface**: Easy-to-use web interface where farmers may input parameters of inputs and see the result.
2. **BACK-END Server:** Processes data, integrates API and connects it with other components.
3. **Machine Learning Module:** Yields crop based on trained models.

The workflow will start with user input that will be handled by the backend system. The machine learning model estimates the forecast of yield using data on the environment and past history. The system subsequently estimates profit based on the prevailing market prices and shows a comparison of various crops.

This strategy allows farmers to consider various alternatives prior to making a cultivation choice.

IV. METHODOLOGY

A. Data Collection

The dataset we have adopted in the research contains different agricultural parameters which are the type of soil, the rainfall, temperature, humidity, and the past yield of the crops. The factors are very important in establishing the yield of crops under varying environmental conditions. Besides these parameters, market price information is also included to determine the economic feasibility and profitability of other crops.

B. Data Preprocessing

The gathered data is pre-treated to enhance the functionality of the machine learning models and reliability. Missing values are also addressed by proper methods and irrelevant or redundant features are eliminated to minimize noise in the data set. The numerical data is standardized to provide uniformity among scales which is useful in improving the accuracy and stability of the models during training.

C. Machine Learning Model.

1.Random Forest:

Random Forest is used in this work as the main model of yield prediction [7]. The algorithm builds an ensemble of decision trees, each of which is trained on a bootstrap sample of the data with a random subset of features possibly used at each split. All trees yield predictions which are averaged to give the final yield estimate. The design minimizes the variance without amplifying the bias and thus the model fits the irregular and region-specifics of agricultural datasets in India.

In this system, the model will be trained based on features that include soil type, annual rainfall (mm), average temperature (o C), relative humidity (percent), and a five-year average historical yield (kg/ha). Estimators will be 200 trees; the maximum depth will be tuned with cross-validation of five folds to achieve a minimum RMSE on the retained validation split. The scores on feature importance calculated during training ensure the two most significant predictors are rainfall and historical yield and the third predictor is soil type, which fits the domain knowledge of the conditions of oilseed cultivation.

The general process of crop prediction and recommendation system is depicted in Fig. 1.

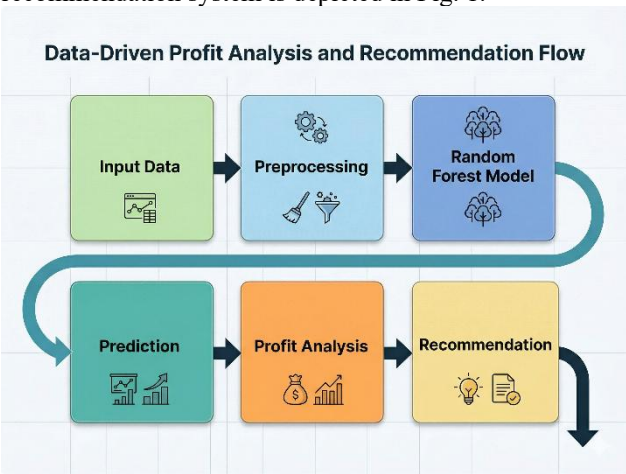

Fig. 1. Crop Yield Prediction and Recommendation Flow

The system works on input data by performing various steps such as preprocessing, feature selection and prediction of yield based on the Random Forest model as illustrated in Fig. 1. The forecasted yield is further evaluated and compared with the prevailing market prices to know the net profitability and the system suggests the appropriate crops according to the maximum profit that has the risk labels.

D. EVALUATION METRICS

Standard measures, like Mean Absolute Error (MAE) and Root Mean Square Error (RMSE), are employed in order to estimate the model performance.

RMSE is defined as:

$$RMSE = \sqrt{(1/N \times \Sigma (Y_i - \hat{Y}_i)^2)}$$

V. SYSTEM IMPLEMENTATION

The suggested system is a full-stack web application that is intended to offer the practical support to farmers. The system combines a user interface (easy to use) and a backend server and a machine learning component to make predictions.

The frontend is created so that farm information like land size, soil type, source of water and season can be easily entered by the user. The interface is made easy and user-friendly such that the user with little technical expertise can be able to use it.

The backend will handle processing user inputs and communicating with the machine learning model. It manages data validation, API communication and generation of results. The machine learning module involves the usage of trained models to forecast crop yield basing on environmental and past information.

The system has various functional modules. These simulators are used to compare the various crops in terms of the yield and anticipated profit. The findings are presented in a graphical form in the form of cards and charts and hence the results are easier to understand by users.

Moreover, the site offers a government schemes section where individuals can find details regarding the subsidies and financial assistance schemes. This assists farmers to make wise choices bearing in mind profitability and benefits at hand.

In general, the application is centered on integrating predictive analytics and a convenient interface to provide an efficient and accessible solution to the target of choosing a crop. In Fig. 2, the implemented crop profit simulation interface is presented.

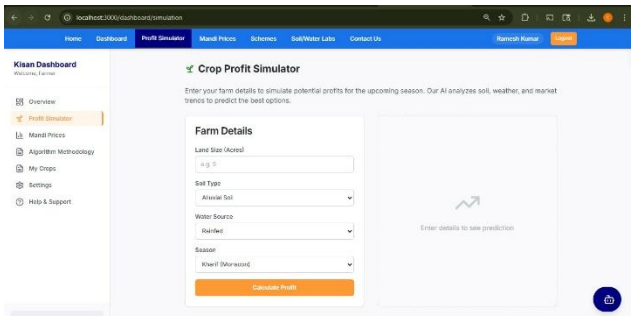

Fig. 2. Crop Profit Simulation Interface

VI. RESULTS AND ANALYSIS

The evaluation of the proposed system was conducted with the help of real-time input parameters by the developed web-based interface. The crop profit simulator module allows users to enter crucial farm information like the size of the land, water source, and soil type, and the time of year when the crop is grown. Using such inputs, the system processes the data and makes predictions of yield, revenue, cost, and net profit of various crops.

To test the system, a sample input of 5 acres of land, alluvial soil type, rainfed water source and the Kharif season was used. The model produced forecasted values of various crops using these parameters thus allowing a comparative analysis of their economic performance.

Table I shows the results of the system.

Crop	Revenue (₹)	Cost (₹)	Net Profit (₹)
Paddy	6,36,761	75,000	5,61,761
Cotton Seed	7,98,742	55,000	7,43,742

Based on Table I, it may be noted that paddy is a safe crop with a comparatively stable returns and cotton seed gives a much higher net profit but is high risk. This classification enables farmers to have an insight not only on profitability but also the level of risk involved in the choice of crops.

Even though the paddy yields steady production, cotton seed has better economic payoff based on the good market prices and relatively low input in comparison to the market revenue. The system presents this difference with the profit comparison interface, which enables people to easily see the most advantageous alternative.

The findings indicate that the use of traditional crops alone might not necessarily maximize income of farmers. Through the proposed system, farmers will be able to consider other crops like oilseeds and make decisions basing on the forecasted profitability, but not assumptions. This fact-based solution promotes diversification and enhanced use of resources.

Moreover, the results are displayed in an intuitive form provided by the visual output of the system in the form of cards and structured displays, where each crop is marked by its level of risk and anticipated profit. This makes it easier to interpret and makes sure that the results are comprehensible to even non-technical users.

The combination of yield forecast and profitability analysis offers a more viable method of operation as opposed to other conventional systems that emphasize on yield estimation. This guarantees that farmers are able to make economically viable decisions as opposed to depending on only the conventional practices.

On the whole, the findings confirm the hypothesis that the suggested system is useful in crop selection through machine learning predictions and financial analysis. This will result in better decision making, less uncertainty and more farmer income potential.

Figure 3 shows the comparison of profits predicted by the system.

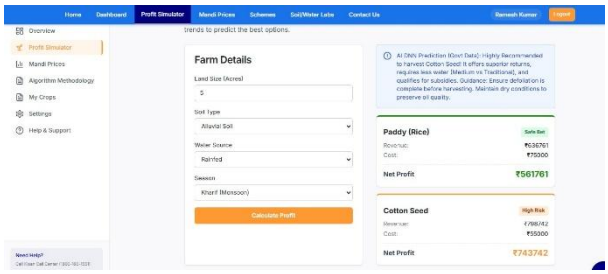

Fig. 3. Crop Profit Prediction Output Showing Revenue, Cost, and Net Profit

VII. CONCLUSION

The current paper has presented a tweaked web based decision support system that can aid farmers to make the correct decision of crops by applying machine learning techniques. The system has many components included and those are: some analysis of environmental parameters, prediction of crop yield and the prediction of profitability so as to provide practical and viable recommendations.

One of the primary contributions of the present work is the combination of predictive analytics with economic evaluation. In the proposed approach, market price and cost analysis are employed in addition to yield prediction, unlike the traditional systems, which are founded on yield prediction of their operation. This will enable the farmers not only to be steered by the output of the product (production), but also the financial result which is one of the highest considerations in actual farming.

The applied model, which is the Random Forest model in this study is effective to explain the complex and non-linear relationships.
crop yield and environmental inputs. Training A cross-validation was carried out to ensure the reliability of the generalisation performance and the system interface presents the predictions in a format that is easily read by the user with very little technical knowledge.

Based on experimental findings, oil seed crops can yield more profit as compared to the traditional crops such as paddy under the appropriate conditions such as cotton seed and groundnut. Besides, the system risk classification option (e.g., safe vs high risk) is more suggestive, and more useful to farmers in order that they can find the proper mix between profitability and stability.

By promoting utilization of data to choose crops, the proposed system will support transitioning into oilseed farming that is vital when it comes to reducing Indian dependence on importation of edible oils. Additionally, the practical value of the available information about the government schemes, due to their increased accessibility, is also beneficial to the practical usefulness of the site since it makes farmers aware of the financial assistance they may claim.

Overall, the system is an efficient, scalable and user-friendly method of modern-day decision making in agriculture. It could boost the earnings of farmers, make agricultural practice sustainable and contribute to the domestic agricultural expansion.

VIII.FUTURE WORK

A small sample crop mix (paddy and cotton seed) at Kharif season with constant historical prices has been used to analyse the present system. To improve the relatability of the platform to the delivery of oilseeds in the objectives of the program to promote the growth of the oilseeds in the state, the crop coverage should include groundnut, sunflower and soybean- the three most popular oilseeds in Karnataka. The current profit module might be linked up with a live mandi price feed, via the Agmarknet API and substitute the existing assumption of price in the existing research and increase the accuracy of the profitability forecasts.

It was trained on aggregated data at the district level; the tendency of this model is that with field-scale responses of IoT sensors to quantify soil moisture and microclimate temperature, the error in prediction is minimized, when commuting smallholder plots with significantly differentiated conditions are involved, which are below the district averages. The change in accuracy would be measured by training the Random Forest using this more finely-grained data, and reverse.

SIH25273 problem statement suggests a progressive web application (PWA) option of the platform that will accept the Kannada and Telugu languages to lessen the literacy gap in the area of Karnataka and Andhra Pradesh targeting farmers [5]. It should have voice input aid and this would again help relieve the burden on those users who lack familiarity in the use of web forms.

Intermodelling: It is possible to envision a time-series LSTM-based extension to be able to model the multi-seasonal yield patterns which cannot be predicted with the existing season-agnostic Random Forest. Comparative runs of the Random Forest control and gradient-boosted models (XGBoost, LightGBM) on the extended oilseed data would assist in perceiving whether a more intricate model would gain any considerable benefits of accuracy to this field.

Internet of Things (IoT) device integration is another improvement possibility. Real-time information about the field can also be obtained using soil moisture, temperature and humidity sensors, allowing more dynamic predictions.

Also, other functionalities can be included into the system such as pest detection system, fertilizer recommendation and irrigation planning. All these would make the platform a complete smart agriculture product.

Overall, the system improvement will make it more efficient, user-friendly, and scalable to make the system

more valuable to farmers and groundbreaking in enhancing the concept of data-driven agriculture.

IX. REFERENCES

[1] Ministry of Agriculture & Farmers Welfare, Government of India, "Agricultural Statistics at a Glance 2023," Directorate of Economics and Statistics, New Delhi, 2023. [Online]. Available: https://eands.dacnet.nic.in

[2] S. Khaki and L. Wang, "Crop Yield Prediction Using Deep Neural Networks," IEEE Access, vol. 7, pp. 123456–123465, 2019.

[3] S. Pudumalar, E. Ramanujam, R. H. Rajashree, C. Kavya, T. Kiruthika, and J. Nisha, "Crop Recommendation System for Precision Agriculture," in Proc. 8th IEEE Int. Conf. on Advanced Computing (ICoAC), Chennai, India, Jan. 2017, pp. 32–36. DOI: 10.1109/ICoAC.2017.7951740.

[4] P. C. Padhan, "Oilseed Production Analysis in India: Trends and Challenges," Journal of Agricultural Economics, 2019.

[5] Smart India Hackathon, "Problem Statement SIH25273 – Agriculture Innovation: Promoting Oilseed Farming," Ministry of Education, Government of India, 2024. [Online]. Available: https://www.sih.gov.in

[6] J. R. Quinlan, "Induction of Decision Trees," Machine Learning Journal, vol. 1, no. 1, pp. 81–106, 1986.

[7] L. Breiman, "Random Forests," Machine Learning Journal, vol. 45, no. 1, pp. 5–32, 2001.

[8] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning, MIT Press, Cambridge, MA, 2016. ISBN: 978-0-262-03561-3.